

Redox reactions

IN A REDUCTION-OXIDATION (REDOX) REACTION, ELECTRONS ARE TRANSFERRED FROM ONE ATOM TO ANOTHER.

SEE ALSO

◀ 112–113	Ionic bonding
Electrochemistry	148–149 ▶
Refining metals	152–153 ▶
Electric currents	203 ▶

A redox reaction is one in which the oxidation state of one reactant rises as the oxidation state of the other falls to balance it. The oxidation state is the number of electrons added to or taken from an atom.

Oxidation states

Chemical reactions occur because most atoms have an incomplete outer shell of electrons, which makes them electrically unstable. To fill their outer shell they form bonds with other atoms in which they accept or donate electrons. The number of electrons that an atom needs to lose or gain to make itself stable is called its oxidation state. Any uncombined element has an oxidation state of zero.

+3									
+2	Cu ²⁺	Zn ²⁺							
+1			Na ⁺						
0	Cu	Zn	Na	0	Cl				
-1					Cl ⁻				
-2				O ²⁻					

△ Positive or negative?

The oxidation state, or number, shows the number of electrons that are gained or lost when an atom changes to an ion (see pages 112–113). The oxidation state of all uncombined elements is zero, as is the sum of the oxidation numbers in a neutral compound. For simple ions, the oxidation state is the same as the electrical charge of the ion.

Changing oxidation state

When atoms or ions undergo a reaction, their oxidation state changes. For example, the oxidation state of sodium changes from 0 to +1 because it has one electron in its outer shell to give away. It is easier for it to donate the electron than to try to fill up its shell with seven more electrons. On the other hand, chlorine has an oxidation state of -1 because it is lacking one electron to complete its outer shell.

sodium has one electron in its outer shell
chlorine lacks one electron in its outer shell

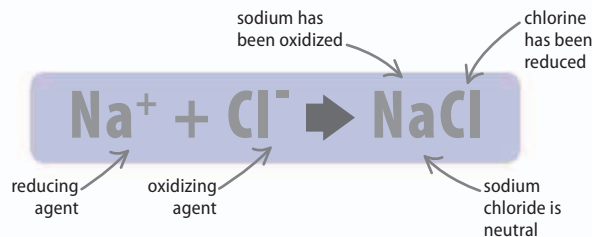


△ Sodium chloride

Sodium and chlorine make an ideal pair to form a compound because sodium has an electron to donate to chlorine.

Oxidation and reduction

When an atom (or ion) loses electrons during a chemical reaction it is said to have been “oxidized.” The term “oxidation” originally applied to reactions where oxygen had combined with another substance, but now it is used in any reaction where electrons are donated. The atom or ion that gains electrons is said to have been “reduced.” All redox reactions happen in pairs—for every reduction reaction there is a corresponding oxidation reaction.



△ Oxidizers and reducers

The atom or ion that accepts the electrons is called the oxidizing agent, oxidant, or oxidizer. The atom or ion that donates the electrons is called the reducing agent.